Date: 16.7.26 Exp: 2

Study of different types of Network cables and Crimping of cables with RJ45 Connector

AIM:

To study the different types of network cable

a) Understanding different types of network cables

The different types of cables used in networking are:

1) unshielded twisted pair (UTP) cable
2) Shielded Twisted pair (STP) cable
3) Coaxial cable
4) Fibre optic cable

Cable type	Category	Max data transmission	Adv/Drs	Application
UTP	Category 3 Category 5 Category 5e	10 bps upto 100 Mbps 1Gbps	→ cheaper → easy to install → more prone to EMI and noise	10 base - Ethernet fast ethernet
STP	Category 6 6a	10Gbps	→ Shielded → faster than UTP → less to noise	Gigabit-Ethernet
Coaxial cable	RG-6 RG-59 RG-11	10 - 100 MBps	→ High band width → versatile → Limited distance → Cost → size is bulky	Speed of signal is 50dm, Television network
Fibre optic Cable	Singlemode, Multimode	100Gbps	→ High Speed → High bandwidth → High security → expensive	Max distance of fibre optics cable is 100 meter

RESULT:

Thus, the different types of network cables were studied and verified.